All shops are not the same, however, and some of the railroads are living right up to the letter of the rules, while others are doing just as little as they can and "get by." If the officers in charge show that they wish to have the rules lived up to, all of the men under them will get busy and do the work, but, should the officers show that they are lax in this respect, the men under them will be lax also.

The way to get all the work done is to have a government inspector for every shop of any size who will say what work shall be done and what work shall not be done, and also see that the work is done. Have him report direct to the district inspector, and receive his pay from the railroad company and make it so that he cannot be removed without an investigation. Then the excuses of no money, no men, short time, small appropriation, or light repairs will be over with. J. A. C.

Rolling Boiler Flues

I read D. G. Young's letter on "Rolling Boiler Flues" with great interest, inasmuch as he states that my method must be "slow." His method is doubtless much faster, and since he has had much more experience, I do not doubt but that he can judge with extreme accuracy just when he has rolled the tube enough. However, it is evident that a novice could hardly tackle the job of rolling flues with no other instructions before him than Mr. Young's description of his method. Experience is the thing that is necessary for rapid work.

I admit that I am dissatisfied with my own method. It *is* slow and it is not absolutely dependable, especially for the apprentice. And since air motors are so much used nowadays, anyway, it is a rather antiquated method.

The thing I would like to get hold of is a *sure* method that can be described in words and would not depend so much upon experience. I can see no reason why a good gage method could not be worked out—a method that would automatically give the diameter of the flue while rolling is in progress. Every flue would thus be given the same internal diameter, the stretch would be the same, the pressure against the tube sheet the same, and the danger of tube sheet deformation would be considerably reduced. I would be pleased to know if there is such an instrument in use, its design, and the service it gives.

Good judgment is all right, but certainty is much better.

New York. N. G. NEAR.

Rolling Boiler Tubes

I have read with much interest and some surprise the communications of Messrs. Near, Mason and Young on the above subject in the December, 1914, and January and February, 1915, issues of THE BOILER MAKER. The kink of which Mr. Near speaks, of feeling the tube, is a very old one, and well known to boiler makers, but the feeling is usually done after the flue is rolled, and the rolls are never removed until the flue is properly rolled, when the boiler maker does the feeling act, not to satisfy himself that the flue is properly rolled, but more from force of habit.

There have been given from time to time in this journal many ways of setting flues, but very few of your contributors have given the secret of good flue setting, *which is the fit of the flue in the flue sheet.* Mr. Young's comment on the removal of the rolls from the flue to feel if the flue has been rolled enough is good, but, as he says, would never be tolerated in an up-to-date shop, but I would like to ask the readers of THE BOILER MAKER what kind of a shop it is that tolerates the insertion of flues in a boiler with an eighth of an inch play between the flue and flue sheet?

To quote his own words: "In most cases the flues have a full eighth of an inch clearance in the flue sheet, and must, of course, stand the strain of being rolled out that much and after that, as much as the boiler maker thinks enough to make it tight." There are many articles in THE BOILER MAKER of recent date bearing on this subject. The editor's comment on a paper written by Mr. Kelly in THE BOILER MAKER, January, 1911, says in part: "No one branch of railroad management can be held responsible for flue failures; the designer of the boiler, the one who is responsible for maintenance, and the operating engineer, all have something to do with it," and I may add, so has the boiler maker. In THE BOILER MAKER for July, 1911, Mr. L. A. Barneman says the boiler maker is the man who has it put up to him, etc. If we look at this subject in the proper light I think that the readers will all agree that Mr. Young has sounded the keynote of flue failures on many, if not all, of our railways, when he says that they tolerate such a fit between the flues and flue sheet on the Southern Railway.

Referring to my letter on flue setting in THE BOILER MAKER, page 397, December, 1913, I will quote a few passages coming down to the present day. I find that there are shops whose practice is very little better than that used forty years ago. Notwithstanding an abundance of modern tools, modern methods are lacking. Again, there is one system of flue setting that has been used by one firm of locomotive builders for the last forty years, which is well worth knowing. This firm, like the early builders, use the straight flue. *The holes in the front flue sheet are cut 1/64 inch large.* For a 2-inch outside diameter flue the hole would be 2 1/64 inches in the front end, while the hole in the firebox flue sheet would be 1/8 larger, or 2 1/8 inches. In the large holes at the firebox end is inserted a copper ferrule 1/16 inch by 3/8 inch by 2 1/8 inch. These ferrules are given a light tap on a mandrel or pin with the hammer. This insures the ferrule taking hold of the flue sheet, while being driven in preparatory to being rolled, which is done lightly enough to allow the flue being driven in place.

During my experience of some thirty-seven years, handling flues of various kinds (brass, copper, iron and steel) and in large quantities, I have made it a practice to always have a tight fit between flues and flue sheets in the new work. The drill press hand was given standard copper ferrules, also a short end of flues of different diameters, and it was his duty to see that his cutter did not cut large and there was no excuse for doing so. Excuses were not tolerated, and on old work the same care was exercised in procuring a tight fit. Where flue holes in the firebox end had been rolled large, copper ferrules of extra heavy wire were used and rolled until the proper fit was obtained. The front end was treated in the same way, all holes that had been cut (or rolled) large for the removal of flues during repairs were ferruled with heavy coppers to get the requisite fit. All other flues were lined (or shimmed up) tight with some soft iron before a roll was applied, so that when the work of expanding was started the work was easily and quickly done, and the tread of the rolls left a slight impression easily felt and understood by the boiler maker.

The secret of good flue work is a good, tight fit, and it is not necessary to drive the pin in the rolls with all a man's strength, for if half our boiler makers realized the enormous pressure exerted by the rolls upon the flue, they would hesitate before doing so. Now I have found it

Letters from Practical Boiler Makers

This Department is Open to All Readers of the Magazine —All Letters Published are Paid for at Regular Rates

Rolling Boiler Tubes

I have followed with great interest the articles on tube and flue setting and rolling, published in previous issues of THE BOILER MAKER, and wish to state that, with the exception of one operation, I agree with the statements made by "Flex Ible" in the April issue. From my experience I have learned that if a tube or flue is rolled after being completely beaded, the bead will expand as well as the tube, therefore causing the *edge* of the finished bead to loosen and creep away considerably from the sheet. In some instances, where brick arches are not applied, the flames in the firebox striking the flue sheet are apt to burn the extending edge of the bead, destroying the bead.

To overcome this trouble I would advise in setting or rolling tubes or flues in locomotive boilers that the tubes should be made a tight fit, as "Flex Ible" stated, but, instead of beading the tube first and then giving it a final rolling, I would apply the tube with a tight fit and expand with sectional expanders, then laying over the bead with a ball hand hammer of sufficient weight to work with ease, or by a compressed air hammer, using a special laying over tool. After this, apply the roller expander and give it two complete turns of the body, holding the rolls in place, and then give the tube a final beading and finish.

By finishing the tube or flue in the above manner, the boiler maker and foreman may feel confident that the bead of the flue is perfectly tight and snug to the flue sheet. This operation does not take a minute longer than that described by "Flex Ible" and, at the same time, it eliminates leaky flues and burned beads, and insures the advantages of roundhouse and hot work. TAD TULIN.

Jersey City, N. J.

Having read many letters in THE BOILER MAKER on rolling boiler tubes, I wish to give the results of my experience.

I never make a practice of putting my finger in a flue to tell whether it is sufficiently rolled or not, as I consider this a waste of time. When I roll a flue in a new tube sheet I always notice that little scales will form on the tube sheet, and, when a tube is rolled tight enough to break these scales off, the flue is rolled hard enough. If the flue is rolled any more, it will be noticed that the end of the flue stretches, which must come from the part of the flue in the tube hole causing unnecessary thinning and weakening of the flue. After watching this action on the first few flues rolled, and watching the expander pin, every flue in the boiler can be properly rolled by letting the pin go in just the same distance on each flue. This can be judged accurately enough with the eye, as I have found working in up-to-date shops and putting in a great many flues, always having good success and making good time on my work.

I think a flue should fit tightly enough in the tube sheet at the back end of a tubular boiler, so that, after it is driven in slightly with a maul, it will require very little rolling. If a sectional expander is used, the flues ought to be rolled slightly before beading. If the flue is beaded and then rolled, you can watch closely and see the bead spring away from the sheet, whereas the bead is supposed to stay tight against the sheet to brace it.

A fault sometimes found in flue expanders is that they feed too rapidly, and when the motor is reversed it releases so suddenly that a small crease is left in the tube just where the rolling was stopped. Others will have so much taper on the pin that in a case where the flue sheet is of much thickness when the outside of the tube is rolled the inside will hardly be tight in the hole.

Bay City, Mich. GLENN LACEY.

As usual, I have read the interesting contributions in the "Practical Letters" department of the April issue of THE BOILER MAKER. One in particular attracted my attention at this time, that by "Flex Ible" entitled "Rolling Boiler Tubes," pages 130-131. I quite agree with the writer of that article that tubes should fit the holes in which they are to be expanded, if satisfactory results are to be expected. I have known cases where tube ends were split in the attempt to expand them in holes that were ⅛ inch larger in diameter than the outside diameter of the tube. As the tube end was ruined, as far as the boiler for which it was intended was concerned, experiments were made to see just what could be done in such a case.

The tube with the split end was taken out and the other end was heated in a charcoal fire until it just began to show red, and then quickly placed in the tube sheet and the expander applied while the end was still hot. But even then the end split just about the time the tube appeared to get tight in the hole. The tube referred to was one of a number that had to be obtained to retube, entire, two boilers. The tubes were bought from a reputable firm and were first class in every respect, and it was not reasonable to expect that they could be expanded into holes that were ⅛ inch larger than the tubes themselves. It may be possible under certain conditions that occasionally a tube may be expanded in a hole that much larger than itself without splitting, but it seems to me little short of criminal carelessness to let such work go.

In the case to which I refer, when it was found that most of the holes had become enlarged from frequent expanding of the old tubes, it was decided not to risk splitting any more than the one tube, which served the purpose of a warning as to what not to do with the rest. So strips of sheet iron a little longer than the circumference of the tubes, and a little wider than the thickness of the headers into which the tube ends were to be expanded (watertube boilers), were made. The ends of these strips were hammered tapered on an anvil so that they could be lapped and maintain an even thickness of metal all the way round the outside of the tubes. One of these strips was placed around each tube end, forming a sleeve or thimble between the tube and the header hole in which it was to be expanded. The thickness of the strips was such that they could be pushed in around the tube ends by hand. When ready, the tube ends, header holes and sheet metal strips were given a coat of thick red lead paint and then expanded in the usual manner without any further trouble. Personally, I know those tubes lasted for many years, and to all intents and purposes seemed to be just as good as though no occasion had arisen for using the home-made ferrules at all. Of course, it goes without saying that it would be better for the tubes to fit the holes at first without the intervention of ferrules. But much better to use ferrules than to risk splitting tube ends, or to risk the job at

all, by expanding tubes into holes that are much larger than the tubes. Rolling tubes is a nice job, and if properly done, and if the boiler is taken care of, they will remain tight for a long while. CHARLES J. MASON.

Scranton, Pa.

After reading in the April issue of THE BOILER MAKER the comments of N. G. Near and "Flex Ible" regarding my letter in the February issue, relative to the rolling of flues, I wish to explain more fully my statements on that subject.

N. G. Near states that no doubt my method is much faster, and that I can judge with extreme accuracy just when the flue has been rolled, but that it is evident a novice could not handle the job of rolling flues with no other instructions before him than what I gave in my letter.

In the first place, no one in the world would expect a novice to be able to handle a job successfully on any kind of work unless he has had some experience. I cannot understand why N. G. Near should think that a man can start right in on a job after reading a little about it in THE BOILER MAKER, or in any other publication. Theory is all right, but in my opinion it is of absolutely no use without practice.

N. G. Near states that he would like to get hold of a sure method that can be described in words and would not depend so much upon experience. It would not be worth while for anybody to serve his time as an apprentice if all we had to do was to carry a book with us and look it over before starting on the job, and then carry out the job according to the directions in the book. It would certainly be a strange sight to have the master mechanic or foreman boiler maker come around and see a man on the job with a book in his hand, and, upon asking what the book was for, to have him say that he had it as a reference in case he forgot how to do the work, as he had had no experience and was obliged to do just as the book told him.

We boiler makers serve our apprenticeship of four or five years, as the case may be, just to learn that "sure" method or a method as sure as education and experience can teach us. Right here on the Southern Railway we are sure when a flue is rolled, but we cannot guarantee it to last for any certain length of time. This is right where the trouble starts, as flues may last for five or six months without a leak and then suddenly start leaking, due in most cases to extra stresses being put on the boiler.

"Flex Ible" says he would like to ask the readers of THE BOILER MAKER what kind of an up-to-date shop it would be that would tolerate the insertion of flues in a boiler with an eighth of an inch clearance between the flue and the flue sheet. Apparently "Flex Ible" has not taken everything into consideration in this matter. In the first place, in making this statement, I was referring to old work, and, in the second place, I was referring to the front end. Readers of the THE BOILER MAKER who are boiler makers will grant me that in the front end an eighth of an inch clearance is not an extraordinarily large amount, as it leaves only one-sixteenth of an inch on each side of the flue all around. In the case of the one or two large holes through which the flues have been taken out, we always put in a copper liner, or shim, either before the flue is put in (rolling it in position then), or by slipping it over the flue after the flue is in position.

Now in the firebox end we always have a neat fit. If we did not, we could not hold our jobs for a day, as any boiler maker knows that the firebox end is where the most particular care must be taken. I do not think that anyone will contradict me when I say that all of the flue troubles from which any railroad suffers are in the firebox, as there are no troubles worth speaking about at the front end.

We have many different sizes of copper ferrules, and when we get ready to put the coppers in the flue sheet we get a size as nearly as possible to what the flues were when they were first put in. Possibly two different sizes will be used, as in the first seven or eight rows from the bottom up the holes may need reaming on account of the fact that the flues have been worked so frequently that it caused the outside edge of the hole to burr up so that it would be impossible to make a very good job unless the holes were reamed out, as the burr on the outside edge would cut the flues when rolling. The boiler maker has a hard job to get the copper ferrules just right, as the ridge around the edge of the hole makes them slip back too far. Now when we put the copper ferrules in the sheet, we do not have them flush with the sheet, but we always set them back one-sixteenth of an inch. When the flue is in position and set the right length for beading, we roll the flues, and in doing so roll the copper out until it is barely flush with the sheet, which is just the way we want it. On the other hand, by leaving the copper flush with the sheet in the first place the copper would stick out around the flue, and not make nearly as good a job, as it would be harder to bead over.

After rolling, we "bell" the flues out with a special tool for that purpose, which we use in a long-stroke hammer. This does away with the old-time method of "lipping" the flues, or turning them over with a hand hammer. After belling the flues out, we prosser them. Some roads designate their prossers by numbers, such as Nos. 1, 2, 3, 4, 5, etc., while others designate them by the size as 1¾ and so on up. For the most part we use the 1¾ rolls and 1¾ prossers in the firebox and 1⅞ rolls in the front end. Where the hole may be a little large, we use the 2-inch rolls. As all our flues are annealed before being put in, they are rolled out very readily, very seldom bursting. If any flues split or burst when the pressure is put on, it is nearly always in the weld, for which the boiler maker is not responsible.

I wish to state to "Flex Ible" that I have rolled many hundred sets of flues with lots of clearance in the front end, and after a couple of years have taken the same sets out, finding that the front end would be in as good condition as the day it was put in. This was not the case with the firebox end, however, which has to stand all the hard work of the flue. In every set of flues that I have taken out in my experience as a boiler maker, I have never seen one taken out on account of leakage at the front end. In my letter published in the February issue, I forgot to mention that after all the coppers are in the firebox flue sheet we use a short end of a flue that has been swagged down until it fits neatly in those coppers. Then all the flues for that boiler are made according to that gage.

On new work, where we renew a front and back flue sheet, we have absolutely no trouble at all, and always have a perfect fit at both ends. I fail to see what excuse could be found in any shop for not having a perfect fit. "Flex Ible" says the secret of good flue work is a good tight fit. I agree with him as far as the firebox is concerned, but it is not necessary to have the flues a tight fit in the front flue sheet in order to make a good job, as the tedious work of lining all flues up in the front flue sheet is something that the railway companies do not wish to stand for, if they can get as good a job in half the time and at half the expense.

"Flex Ible" tells us not to drive the pin in the rolls with all a man's strength. Whoever heard of a boiler maker

matic reverser. On new work I should think this could be applied so well that a second rolling would be unnecessary. Mr. Williams says: "If some device were put on the expander pin so that when the pin has traveled just so far it will reverse the machine, it would be far better if one or two tubes leak on the test. They could afterwards be rolled tight, and it would be far better than stretching the tube sheet and the tube, which is very often done even by good men on tubes over 2½ inches diameter."

Flex Ible's arguments anent sizes of tubes and stretch necessary to make them fit their holes are certainly based on sound logic. If it would be possible to make tubes in such a way that they would need no stretching on installation, such tubes would be a godsend, providing the cost of installation would not be much more than the present cost of installing tubes that must be rolled or prossered.

In structural iron work, as in the construction of bridges, frames for steel buildings, etc., girders and columns are condemned that have been bent cold to the extent that the elastic limit has been passed. That is, as soon as a piece intended for the structure has been permanently deformed it will not be used.

Tubes rolled into boilers are purposely deformed, as it is the only method, to date, that is practical. Therefore, I argue, if deformation is necessary although deleterious, let us make the degree of deformation as small as possible. The closer the fit, then, before rolling, the better.

Again, in support of Flex Ible's arguments, I might point out that the steel used in structural steel work can also be bent, twisted and battered in a thousand shapes without evident harm, just as is the case with boiler tubes. Yet, as I have said, they won't allow a bit of it in actual practice.

I am pleased to note that the discussion on rolling tubes has reached the proportions it has. It has been most interesting and educative to me.

New York. N. G. NEAR.

Working Old Flues

As Mr. Near wants to hear from the boiler makers on the subject of working old flues, I will give my ideas and show just how I work it when I have to have a helper on the job with me instead of a boiler maker.

A man with experience is the best on old work, for your experience is the best judgment on how to tell when an old flue ought to be tight. Mr. Charles J. Mason is right when he says an experienced man can roll a flue with his eyes shut. You could give an experienced man a self-feeding expander with a ratchet handle and he could roll a whole set of tubes blindfolded and make a good job of them.

The whole idea on flues is speed. You must handle the job to keep yourself and helper busy and do the work quickly. When the foreman starts a boiler maker on a flue job, the next thing he tells you is that the inspector will be here to-day or to-morrow to test the boiler; also they say that some other boiler maker did a job just like this in such a time and you must do as well or better, so it takes experience to get the idea when a flue is properly rolled and speed and head work to finish in a reasonable length of time. Time is the whole thing on the job after you know how the work ought to be done.

When retubing an old boiler, the boiler maker generally finds the flue holes in the back head to be of different sizes. On account of the old flues being rolled at different times, some are rolled more than others, according to where they happen to leak. Each time these flues are expanded the tube holes are also expanded a little. Now to make a good job of the flues and to make fair time I find it a good policy to put in copper ferrules, and by using ferrules of different gages to fill the flue holes I make the tube holes very nearly a uniform size, which gives the flue a neat fit in the hole. The ferrules can be put in the boiler and then a slight roll with an expander will hold them in place. The boiler maker can start to put in the ferrules, and as soon as he has a few in to make room for his helper or apprentice the helper can start rolling them. As it takes but very little rolling, the helper will be able to keep the boiler maker moving to keep ahead of him. An apprentice or a helper that has some practice at the work can put these ferrules in alone after being shown how to do it on two or three tube holes.

After these ferrules have been set and the tubes put in place ready for rolling the boiler maker can take the expander and roll one or two flues to get an idea of about how far to let the expander go in. Then he can hand it to the helper and let him get to work with it, showing him how to work it for a few flues and about the distance to let the pin go in. The helper will be able to roll the flues while the boiler maker is cutting off the flues on the front end and doing the beading.

If it is not desirable to put in the copper ferrules, you can put the flues in place and take some soft iron strips and cut them in pieces long enough to fill in around the flue and scarf each end down even and thin so the expander rolls will roll over without blocking. The helper can get this packing ready while the boiler maker is packing the flues. After the flues are all packed you can give the helper the same instruction as when putting the ferrules in place and finish the job in the same manner.

I have used these methods when out on repair jobs, where I had to use a helper instead of a boiler maker, and I have found them to give satisfaction in time on the job and my work tested satisfactorily. When I roll flues on old boilers, either by hand or by machine, I roll them just enough so that it makes the flue solid and smooth. I do not go to extremes by rolling a flue very hard, because that is only waste of time, besides weakening the flue. I do not expect a set of flues to stand the cold-water test without having a few flues sweat a little. I would rather roll a few flues when the test is on than to go to extremes, and I find it the cheapest method. I have done several jobs where I have had perfect tests and other jobs where I might have five or six flues to roll a little after testing.

Bay City, Mich. GLENN LACEY.

Talks to Young Boiler Makers

A boiler under 250 pounds pressure had not let go in the office, but the noise that was coming out of the door could be compared to it. What was making the noise was a man. In one hand he held a document ornamented with a nice little blue ribbon sewed in and out through eyelet holes. There was an engraving of the U. S. Patent Office right near the top and the man's name was beautifully written below it. In his other hand was a nice morocco case with his name in gold letters across the top, and in much larger letters the name and address of some patent attorneys.

About a year before the man had received in the mail this nice case with the patent that the United States had granted him inside it. He felt pretty sure that now he had the world by the tail and he was a very proud citizen. He was very grateful to the patent attorney for placing a fortune in his hand. The noise he was now making was cussing out the government of the United States, the